Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Helium nuclei / 2 protons and 2 neutrons (1) accept ${}^{4}_{2}\mathrm{He}$ Emitted from the nucleus [at high speeds] (1)	2			2		
	(b)			$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{432} = 1.60 \times 10^{-3}\ \mathrm{yr}^{-1}$ **or** $5.08 \times 10^{-11}\ \mathrm{s}^{-1}$ (1) **Either:** Substitution: $A = A_0 \exp\left(-\left(1.60 \times 10^{-3}\right)(30)\right)$ (1) $\frac{A}{A_0} 100\% = 95.3[\%]$ (1) $\%\ \text{decrease in activity} = 100 - 95.3 = 4.7[\%]$ (1) **or:** 30 years << ½-life [or by impl.] (1) $\therefore$ Fractional decay in 30 years = 30λ = 0.048 (1) $\therefore$ % decrease in activity = 4.8[%] (1) Accept 5% **Alternative:** 30 years is 30/432 = 0.07 half-lives (1) A = $½^{0.07}$ (1) = 0.95 (1) So activity has decreased by 5[%] (1)	1	1 1 1		4	3	
	(c)			**Any 2 × (1) from:** • Alpha particles pose little risk unless inhaled or ingested. • Very few (or none) make it to the outside of the detector. • If they do, they are stopped by a few cm of air and are unable to penetrate surface of skin. **To award both marks there must be a conclusion present i.e. a minimum of no stated.**			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			**Indicative content:** <u>Absorbers:</u> Explain how absorbers can be used to distinguish between the three radiations – describe set up. Diagram or a clear explanation. Sheet of paper stops alpha. Few mm of aluminium will stop alpha and beta. Gamma penetrates both the paper and aluminium. Reference to background radiation. <u>Magnetic field / Electric field</u>: Explain how a magnetic / electric field can be used to distinguish between the three radiations – describe set up. Diagram and clear explanation. Deflection of alpha particles much smaller than for beta, charge positive. Deflection of beta particles opposite to that of alpha, as charge-negative. No deflection of gamma. Correct direction of alpha or beta in magnetic / electric field. **5-6 marks** Absorbers and magnetic / electric field both covered comprehensively. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** Either absorbers or magnetic / electric field covered comprehensively or limited account of both areas. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.*	4	2		6		6

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Limited account of either absorbers or magnetic / electric field. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						
				Question 3 total	**7**	**5**	**2**	**14**	**3**	**6**